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An Expert-Annotation-Guided Sampling and Adaptive Soft-Label Fusion Method for Global Land-Cover Segmentation From Remote Sensing Images

Jiaqi Xiong, Chenxiao Zhang

***Abstract*—Global land-cover segmentation using remote sensing images requires a large amount of training data collected from experts and non-experts. This mixed-source annotation process introduces three major challenges for model training: class imbalance, geographic imbalance, and label noise from non-expert annotations. While class imbalance has been extensively explored in the remote sensing field, the latter two have received less attention, especially in the context of global land-cover mapping with mixed-quality annotations. In this paper, we propose an expert-annotation-guided sampling and adaptive soft-label fusion method for global land-cover segmentation with annotations from experts and non-experts. First, a teacher model trained solely on expert annotations is constructed to provide supervision signals for the non-expert annotations. Second, we construct a guided sampling prior over the entire training set based on category difficulty, geographic under-coverage, and teacher-estimated sample quality. At each training iteration, this guided prior is mixed with a uniform distribution to draw the mini-batch, increasing the sampling rate of geographically under-covered regions, difficult categories, and reliable samples while limiting the distribution shift caused by aggressive guided sampling. In parallel, adaptive teacher-guided soft-label fusion is applied to non-expert annotations, with the fusion weight determined by teacher confidence and teacher-label agreement. We evaluate the proposed method with eleven backbones on the Dynamic World dataset. In the three-seed DOFA analysis, the uniform-sampling baseline achieves $(49.21 \pm 1.04)\%$ mIoU, while the combination of conservative guided sampling and adaptive soft-label fusion reaches $(50.97 \pm 1.03)\%$. Guided sampling alone changes mean mIoU by -0.28 percentage points, but contributes an additional $+0.85$ percentage points under adaptive soft-label supervision, corresponding to a positive descriptive interaction of $+1.13$ percentage points. These results indicate that adaptive soft-label supervision provides the more consistent improvement, while guided sampling offers a conditional average benefit rather than a reliable standalone gain.**

***Index Terms*—Land cover segmentation, Mixed annotations, Geography-aware sampling, Teacher-guided training, Soft-label fusion**

I. Introduction

Remote sensing semantic segmentation plays an important role in global land-cover mapping, environmental monitoring, and agricultural resource investigation [1], [2]. The recent decade has witnessed a methodological shift from feature-based to deep learning-based approaches [3]. Early efforts focused more on architecture design and model development under regional evaluation settings. More recently, the remote sensing field has shifted from regional mapping toward global mapping, mainly due to the success of large foundation models and the availability of large-scale data [4].

In large-scale mapping, high-quality labels are expensive and difficult to collect. Current approaches often rely on a mixture of expert and non-expert annotations [4]. Expert labels are usually more reliable, but they are limited in quantity. Non-expert labels are more abundant and cost-effective, but their quality is uneven and typically less reliable than expert labels. This mixed annotation setting brings three critical challenges for model training: class imbalance, geographic imbalance, and label noise from non-expert annotations. Specifically, class imbalance makes rare categories hard to learn [5]–[8]. Geographic imbalance makes under-covered regions less represented. Moreover, the uneven quality of non-expert labels brings label noise, which can mislead the model.

Existing efforts mainly focus on either sample selection or label correction. Re-weighting and re-sampling methods reduce class imbalance, but they are not able to handle the noisy-label problem [5]–[8]. Other efforts, such as semi-supervised and noisy-label methods, use pseudo-labels, consistency, sample selection, or teacher guidance to improve weak supervision [9]–[13]. While these efforts can alleviate noisy-label issues, geographic imbalance is less explicitly considered. This gap motivates a method that jointly considers class difficulty, geographic under-coverage, and non-expert label reliability.

In this paper, we propose an expert-annotation-guided sampling and adaptive soft-label fusion method for global land-cover segmentation with annotations from experts and non-experts. First, a teacher model trained solely on expert annotations is constructed to provide supervision signals for the non-expert annotations. Second, we construct a guided sampling prior over the entire training set based on category difficulty, geographic under-coverage, and teacher-estimated sample quality. Finally, at each training iteration, we mix this guided prior with a uniform distribution to draw the mini-batch, which increases the sampling rate of geographically under-covered regions, difficult categories, and reliable samples, while mitigating the training distribution shift caused by aggressive guided sampling. In parallel, adaptive teacher-guided soft-label fusion is applied to non-expert annotations, with the fusion weight determined by teacher confidence and teacher-label agreement.

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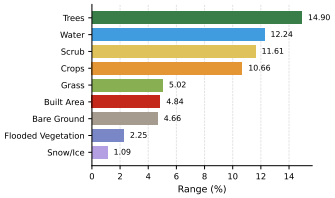
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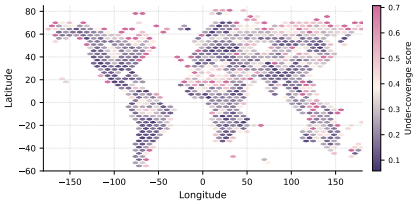
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Class	Pixel ratio (%)	Baseline IoU (%)	Diagnostic
Grass	2.17	15.47	rare + difficult
Bare Ground	3.11	21.19	rare + difficult
Flooded Vegetation	2.18	30.13	rare + difficult
Scrub	30.85	32.72	frequent but difficult
Crops	11.71	48.57	frequent + easier
Snow/Ice	2.73	61.06	rare but easier
Water	7.75	61.71	frequent + easier
Trees	35.29	61.82	frequent + easier
Built Area	4.21	70.27	frequent + easier

(a) Class-wise rarity and reference IoU.



(b) Annotation ambiguity across categories.



(c) Global geographic under-coverage.

Fig. 1. Empirical motivation. Class frequency and reference IoU are not aligned across the nine valid classes; annotation ambiguity varies across categories; and training samples are geographically uneven. These patterns motivate class-aware sampling, reliability-aware label use, and geographic coverage control.

Matched-seed experiments on DOFA indicate that guided sampling alone does not consistently improve segmentation accuracy. Its additional average benefit becomes more apparent when combined with adaptive soft-label supervision, yielding a positive descriptive sampling–supervision interaction of 1.13 percentage points. This observation motivates interpreting the proposed approach as a coupled training strategy rather than attributing its gains to guided sampling alone.

The main contributions of this work are as follows:

1) We identify three critical challenges in global land-cover segmentation with mixed expert and non-expert annotations: class imbalance, geographic imbalance, and label noise from non-expert annotations. These challenges motivate jointly considering training exposure and supervision reliability.
2) We propose a teacher-guided training method that addresses these challenges through two components: a class-, geography-, and quality-aware guided sampling distribution mixed with uniform sampling, and adaptive teacher-guided soft-label fusion for non-expert annotations.
3) We evaluate the proposed method with eleven backbones on the Dynamic World dataset and controlled ablations, including a matched three-seed DOFA study. These experiments show that guided sampling provides a conditional additional average benefit under adaptive soft-label supervision, supporting the two modules as a coupled training strategy rather than attributing the gains to guided sampling alone.

II. Related Work

To position our work, we organize the related studies into three parts. First, we review progress in global land-cover mapping and remote sensing segmentation, including land-cover products, annotation data, and segmentation models. Second, we discuss remote sensing foundation models. These models provide stronger representations for land-cover segmentation, but they still require downstream fine-tuning. During fine-tuning, the model still faces sample selection, class imbalance, geographic imbalance, and inconsistent annotations. Third, we discuss training under imbalanced and noisy supervision. Existing methods address parts of this problem, but they do not jointly consider class difficulty, geographic coverage, and non-expert label reliability.

A. Global Land-Cover Mapping and Remote Sensing Segmentation

Global land-cover mapping is a basic task in remote sensing. It supports environmental monitoring, urban analysis, agricultural management, and climate studies. Many regional and global land-cover products have been developed for this purpose [3], [4], [14], [15]. These products use satellite imagery, expert rules, classical classifiers, or deep learning models to describe land cover at large scale. Dynamic World further improves the temporal coverage of land-cover mapping and provides near-real-time land-cover products [4]. This makes global land-cover segmentation an important setting for studying large-scale remote sensing models.

From the model side, earlier remote sensing analysis often relied on pixel-based classification and object-based image analysis. In these methods, segmentation is used to define image objects before feature engineering and classification [16], [17]. However, their performance often depends on segmentation scale, feature design, and dataset-specific parameters [16], [17].

With the development of deep learning, dense prediction models have become common choices for semantic segmentation. Foundational architectures such as FCN, U-Net, and DeepLab have shaped many later segmentation models [18]–[20]. They have also been widely used for land-cover segmentation in remote sensing images. Recent reviews show that deep segmentation models have improved accuracy on high-resolution imagery, complex boundaries, and multi-class land-cover scenes [3], [14], [15], [21].

However, much of this work focuses on land-cover product generation, network design, and prediction accuracy [3], [14], [15], [21]. For large-scale global mapping, the training data itself can also become a major problem. Different land-cover classes can have very different sample frequencies. Some classes appear mainly in specific ecological regions or specific map tiles. Large-scale datasets may also contain labels from different sources, including expert annotations, non-expert annotations, and labels derived from existing products. As a result, label quality is not always the same.

Therefore, the problem is not only which model should be used for prediction. It is also how the training data should be organized and how mixed-quality supervision should be used. This motivates the need for stronger representations and better downstream training strategies.

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TABLE I
POSITION OF THE PROPOSED FRAMEWORK WITH RESPECT TO THE MAIN RELATED RESEARCH LINES.

Category	Main focus	Gap with respect to this work
Global land-cover mapping and remote sensing segmentation [3], [4], [14]–[21]	Land-cover products, segmentation models, and prediction accuracy	Limited attention to mixed-quality supervision, geographic unevenness, and training-data organization
Remote sensing foundation models and downstream fine-tuning [22]–[32]	Strong pretrained representations and transfer ability	Still require target-task fine-tuning and remain affected by sample imbalance and label noise
Training under imbalanced and noisy supervision [5]–[13], [33]–[40]	Class balancing, sample weighting, teacher guidance, and robust supervision	Usually handles sample exposure and label reliability separately
Proposed framework	Class-, geography-, and quality-aware mixed sampling + adaptive teacher-guided soft-label fusion	Jointly considers class difficulty, class-region under-coverage, and non-expert label reliability

B. Remote Sensing Foundation Models and Downstream Fine-Tuning

Recent remote sensing foundation models use large-scale pretraining to improve transfer across regions, sensors, and tasks. Models such as DOFA [22], Prithvi-EO-2.0 [23], TerraFM [24], THOR [25], Copernicus-FM [26], SMARTIES [27], and SelectiveMAE [28] learn general representations from large remote sensing datasets. General vision, spectral, and vision-language models such as ConvNeXtV2 [29], SigLIP2 [30], CARL [31], and GeoRSCLIP [32] also provide useful backbone choices for remote sensing tasks.

These models address part of the limitation of earlier land-cover segmentation models. They provide stronger feature representations while reducing the need to train from scratch. They can also improve accuracy in new regions or downstream tasks. This is important for global land-cover mapping, where the target regions are diverse and the land-cover patterns are complex. In this sense, foundation models provide a stronger starting point for remote sensing segmentation.

However, foundation models do not remove the need for downstream fine-tuning. A strong pretrained backbone still needs to be adapted to the target dataset and label system. During this process, the model faces the training-data problems discussed above. Some classes are under-represented. Some class-region combinations are weakly covered. Some labels are noisy or ambiguous, especially when non-expert annotations are used.

This means that foundation models mainly improve representation learning, but they do not fully solve the problem of supervision quality. They also do not decide which samples should be used more often during fine-tuning. Different backbones may respond differently to the same training strategy because their features, calibration, and noise sensitivity are different. Therefore, the key question moves from only selecting a strong backbone to designing a fine-tuning strategy that can use imbalanced and mixed-quality data more carefully.

C. Training under Imbalanced and Noisy Supervision

During downstream fine-tuning, re-weighting, re-sampling, class-balanced or focal losses, and logit adjustment reduce the dominance of frequent classes [5]–[8]. They control global class exposure, but do not by themselves describe class-region coverage or whether a dense target is reliable. In global land-cover data, a class can be common overall yet poorly represented in particular biomes, so class frequency alone is an incomplete training priority.

Weakly supervised remote-sensing methods use image-level or coarse labels, CAM refinement, pseudo-mask generation, or cross-teaching to reconstruct denser supervision; SAM-guided ATSG and SGCT-Net are examples of this line of work [9], [13], [36]–[40]. Our data already contain dense expert and non-expert annotations, so the central distinction is not sparse-to-dense mask generation: it is how to allocate exposure and how strongly to correct a non-expert pixel target. Remote-sensing foundation-model adaptation changes the representation available to the downstream head, but does not automatically remove class-region imbalance or label unreliability.

Cross-modal noisy-label studies provide a complementary comparison. CromSS uses cross-modal pretraining and sample selection under noisy labels for segmentation [41], while Difference-Complementary Learning uses optical–SAR consistency and pseudo-label reassignment in multimodal semi-supervised segmentation [42]. Those methods address cross-modal agreement and noisy or incomplete supervision; this paper uses six-band optical inputs with an expert-only teacher and mixed dense annotations, without a second sensor or a cross-modal consistency loss. We therefore combine class-, geography-, and quality-aware exposure with adaptive pixel-wise fusion, while treating the mechanisms in these related settings as distinct rather than interchangeable.

III. METHOD

A. Empirical motivation.

Before introducing the proposed framework, we examine class-wise difficulty, geographic coverage, and supervision reliability in the Dynamic World setting. Figure 1 combines training-data class distributions, geographic coverage, and annotation reliability statistics with Google Dynamic World reference IoUs on the test set to motivate the proposed design. As summarized in Fig. 1, these observations suggest that the challenge of the current task cannot be explained by class frequency alone.

First, class difficulty is not the same as class rarity. Figure 1(a) shows that Grass is rare, with a 2.17% pixel ratio and only 15.47% baseline IoU. Bare Ground is also difficult,

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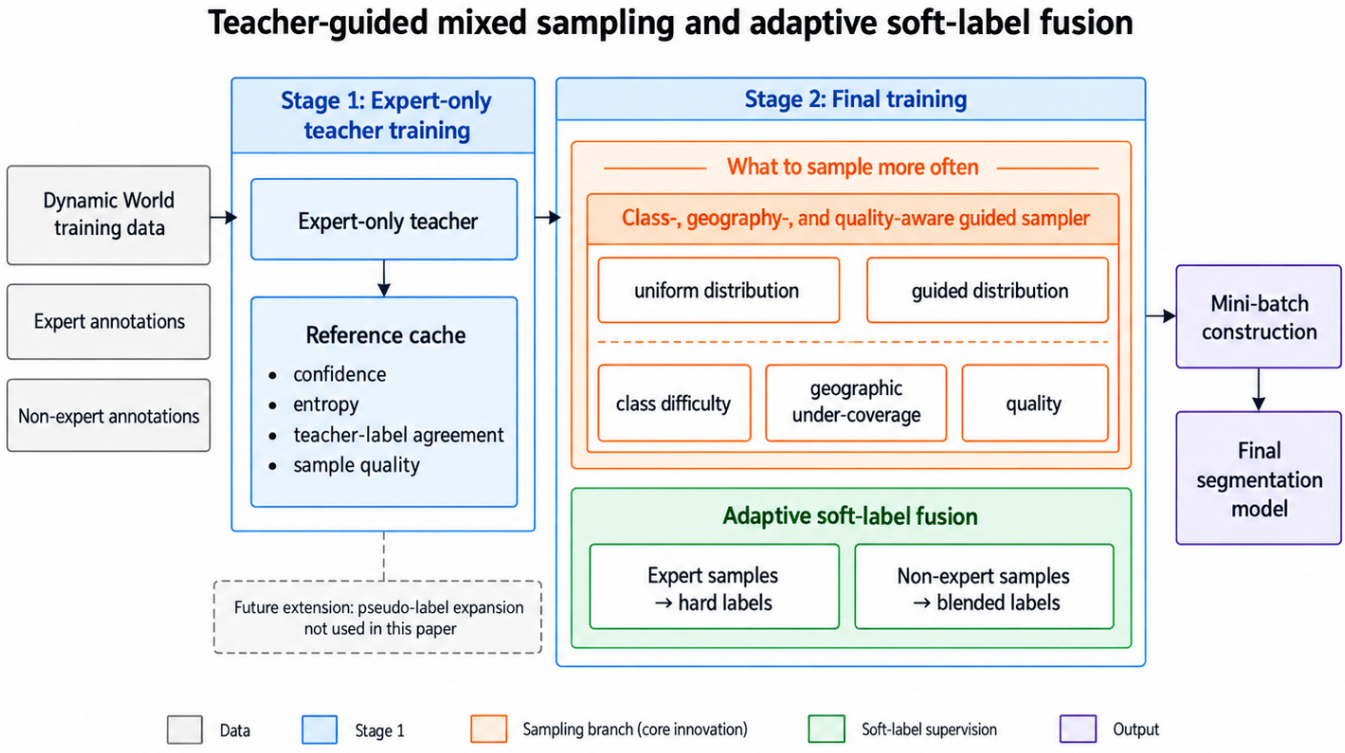


Fig. 2. Overview of the proposed teacher-guided training framework. For each backbone, an expert-only teacher is first trained from expert annotations. The teacher is then applied to the full training set to build a reference cache of sample-level entropy, teacher-label agreement, and quality statistics derived from teacher confidence and label agreement; pixel-wise teacher probabilities are computed online during student training. During final training, the sampling branch mixes a uniform distribution with a class-, geography-, and quality-aware guided distribution. The supervision branch applies adaptive teacher-guided soft-label fusion to non-expert samples, while expert samples keep hard labels. The gray dashed pseudo-label expansion branch is shown only as a future extension and is not used in the experiments reported in this paper.

with a 3.11% pixel ratio and 21.19% baseline IoU. Flooded Vegetation is rare and difficult, with a 2.18% pixel ratio and 30.13% baseline IoU. In contrast, Scrub is frequent, with a 30.85% pixel ratio, but its baseline IoU is still only 32.72%. Snow/Ice is rare, with a 2.73% pixel ratio, but its baseline IoU reaches 61.06%. Built Area has a 4.21% pixel ratio and reaches 70.27% baseline IoU. These examples show that frequency alone cannot determine sampling priority. This motivates class-aware sampling based on empirical difficulty rather than pixel count alone.

Second, the training samples provide uneven geographic support. As shown in Figure 1(c), some regions are well covered, while others are weakly represented. This does not mean that the true global land-cover distribution should be uniform. Rather, the problem is that some class–region combinations receive much less training exposure than others. We define these combinations using the class label together with the biome and MGRS tile of each sample. Geography-aware sampling is therefore used to increase the exposure of under-covered class–region combinations.

Third, supervision reliability is also variable across categories. Figure 1(b) shows that annotation ambiguity differs across classes. Water, Trees, Crops, and Scrub have high ambiguity, while Flooded Vegetation and Snow/Ice are lower. Grass, Built Area, and Bare Ground are in the middle. This means non-expert labels should not be trusted equally across classes. It motivates adaptive teacher-guided soft-label fusion instead of treating all non-expert labels in the same way.

Taken together, these observations suggest that the key challenge of the current task is not simply class imbalance, but the joint effect of class-wise difficulty, geographic under-coverage, and supervision reliability. This empirical diagnosis motivates a framework that combines conservative mixed sampling with adaptive teacher-guided soft-label fusion.

B. Overview

Remote sensing semantic segmentation under Dynamic World-style annotation faces three coupled challenges: class imbalance, geographic imbalance, and non-expert label noise. We address these challenges with a teacher-guided training pipeline. Sample exposure and supervision reliability are related but distinct controls: the sampler changes which patches enter training more often, while fusion changes the target assigned to valid non-expert pixels. We instantiate the pipeline separately for each backbone in our cross-backbone evaluation.

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For each backbone, we first train an expert-only teacher using only expert annotations. We then apply the frozen teacher to the full training set and build a frozen reference cache. The frozen cache stores sample-level entropy, agreement, and quality statistics, with quality derived from teacher confidence and label agreement. Pixel-wise teacher probabilities are computed online by the frozen teacher during student training and are not stored as full-resolution maps. Final training uses both expert and non-expert samples. The sampling branch controls which samples are seen more often (exposure control), whereas the supervision branch controls how strongly non-expert labels are trusted (supervision control).

Figure 2 gives a schematic view of the training pipeline. The figure distinguishes the sample-level frozen cache from pixel-wise teacher probabilities, which are computed online during student training.

C. Expert-only Teacher Training

Let x denote an input remote sensing patch and y its pixel-wise label map. Let $\mathcal{C} = \{1, \ldots, C\}$ be the set of valid semantic classes, excluding the ignore class. Let Ω_{valid} denote the set of valid pixels. The training set is divided into an expert subset $\mathcal{D}_e$ and a non-expert subset $\mathcal{D}_n$.

For each backbone, we train a separate teacher using only $\mathcal{D}_e$. This design keeps the teacher anchored to the more reliable labels. It also avoids treating one backbone as the default teacher for all experiments.

To reduce the effect of class imbalance, the teacher is trained with a class-weighted, logit-adjusted objective. Let z_u denote the logits at pixel u. For each training dataset used to fit a model, class-area ratios are summed over its loaded patches, excluding class 0 (ignore), to obtain the valid-class priors:

$$\pi_c = \frac{\text{counts}_c}{\sum_{k=1}^{9} \text{counts}_k}, \qquad c = 1, \ldots, 9. \tag{1}$$

The teacher and student compute these quantities separately from their respective expert-only and train-all datasets. For valid classes, weights are

$$w_c = \text{clip}\left(\frac{1/\pi_c}{\min_{k=1}^{9}(1/\pi_k)}, 1, 10 \right), \quad c \in \mathcal{C}. \tag{2}$$

Here, $w_0 = 0$ and the implementation sets $\pi_0 = 1$ for finite logit adjustment. With $\tau = 1$, training logits are adjusted as

$$z'_{u,c} = z_{u,c} - \tau \log \pi_c. \tag{3}$$

The teacher objective is the class-weighted cross-entropy over valid pixels, averaged by their count:

$$\mathcal{L}_{\text{teacher}} = -\frac{1}{|\Omega_{\text{valid}}|} \sum_{u \in \Omega_{\text{valid}}} w_{y_u} \log \text{softmax}(z'_u)_{y_u}. \tag{4}$$

The trained teacher is frozen after this stage and is later used to score samples and provide soft targets. No teacher drift or online teacher update is used during student training. The adjusted logits are used only in the training loss; cache statistics, online teacher probabilities, and student inference use raw logits.

D. Reference Cache Construction

After teacher training, we apply the teacher to the full training set $\mathcal{D}_{\text{train_all}} = \mathcal{D}_e \cup \mathcal{D}_n$. For each training sample x_i, the frozen expert-only teacher provides sample-level reference statistics for sampling. The cache stores entropy, teacher-label agreement, and a quality score combining teacher confidence and agreement. The full pixel-wise teacher softmax map is not cached; the frozen teacher produces these probabilities online during student training.

Let $p^T_{i,u}$ denote the raw-logit teacher softmax distribution over all ten classes, including class 0 (ignore). For sample-level quality estimation, $\gamma_{i,u} = \max_{c \in \{0\} \cup \mathcal{C}} p^T_{i,u,c}$ is computed from this distribution at each valid pixel u. Let

$$h_{i,u} = -\frac{1}{\log 10} \sum_{c \in \{0\} \cup \mathcal{C}} p^T_{i,u,c} \log p^T_{i,u,c} \tag{5}$$

denote the teacher entropy. Let $\hat{y}^T_{i,u} = \arg\max_{c \in \{0\} \cup \mathcal{C}} p^T_{i,u,c}$ denote the teacher-predicted class. We define teacher-label agreement as

$$a_i = \frac{1}{|\Omega_{\text{valid}}|} \sum_{u \in \Omega_{\text{valid}}} \mathbb{I}\left[\hat{y}^T_{i,u} = y_{i,u}\right]. \tag{6}$$

Here, the indicator function equals one when the teacher-predicted class matches the reference label at a valid pixel, and zero otherwise. We further define the sample-level quality score as

$$q_i = \frac{1}{2} q_i^{\text{conf}} + \frac{1}{2} a_i \tag{7}$$

where q_i^{conf} is the mean of the raw ten-class confidence $\gamma_{i,u}$ over valid pixels. For non-expert samples, the confidence–agreement combination is a fixed empirical setting, and the resulting score is clipped to $[0.05, 1]$; expert samples use the trusted-anchor score $q_i = 1$. The cache contains sample-level summaries rather than full pixel-wise probability maps. During student training, the frozen teacher generates pixel-wise soft targets online. Neither the teacher nor the reference statistics are updated during this stage.

E. Guided Sampling and Conservative Mixed Sampling

We use the teacher cache and class–region statistics to construct a guided sampling prior. The guided score combines class-specific training priority, geographic under-coverage, and teacher-estimated sample quality. For class c in sample x_i, let $r_c(x_i)$ denote the area ratio of that class, and let α_c denote a fixed empirical class-difficulty coefficient.

To account for geographic coverage, let $b(x_i)$ and $m(x_i)$ denote the biome and MGRS tile of sample x_i, respectively. We define the class-specific geographic factor as

$$g_c(x_i) = \frac{1}{\sqrt{N(c, b(x_i), m(x_i)) + 1}} \tag{8}$$

where $N(c, b, m)$ counts training patches in which the area ratio of class c exceeds 1% for biome b and MGRS tile m. If the MGRS identifier is unavailable, it is replaced by the token UNK, and the additive $+1$ provides smoothing. The dataset-provided MGRS tile identifier is used as a reproducible

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practical spatial grouping rather than as a tuned geographic-resolution hyperparameter. Combining MGRS with biome retains both spatial and ecological context without introducing an additional data-dependent geographic partition. The 1% area-ratio threshold is used only to define meaningful class presence in a patch and is not tuned as an accuracy-optimization parameter. Thus, $g_c(x_i)$ assigns a larger value to class–region combinations that are less represented in the training set.

For the geographic diagnostic, we aggregate the class-specific occurrence counts into a patch-level coverage score. Let $\mathcal{C}_i = \{c : r_c(x_i) > 0.01\}$ denote the valid classes with meaningful area in patch x_i. We define

$$s_i^{\mathrm{cov}} = \frac{\sum_{c\in\mathcal{C}_i} r_c(x_i) N(c, b(x_i), m(x_i))}{\sum_{c\in\mathcal{C}_i} r_c(x_i)}. \tag{9}$$

This diagnostic is the area-ratio-weighted training support of the meaningful classes present in a patch; it uses the original training-derived counts $N(c, b, m)$ and is separate from the class-difficulty and sample-quality terms used by the sampler. We set $T_{\mathrm{cov}} = \mathrm{median}\{s_i^{\mathrm{cov}} : x_i \in \mathcal{D}_{\mathrm{train}}\} = 3.41$. A patch is under-covered when $s_i^{\mathrm{cov}} \le T_{\mathrm{cov}}$ and well-covered otherwise. Both training and test patches are scored using the training-derived counts and this fixed training-set threshold; no test-set median is computed.

Using the cached sample-quality score q_i defined above, we combine these terms to obtain the guided score of sample x_i:

$$w_i = \left(\sum_{c\in\mathcal{C}} \alpha_c\, r_c(x_i)\, g_c(x_i)\right) q_i \tag{10}$$

The class-wise summation aggregates priority according to class content, empirical difficulty, and geographic under-coverage, while q_i modulates the resulting sample-level score according to teacher-estimated sample quality. The coefficients α_c are a manually assigned empirical difficulty prior informed by validation class-wise performance observations. They were not computed by transforming Google Dynamic World test-set IoUs; the test-side class scores reported in the motivation figure are descriptive evidence only. For the class order ignore, water, trees, grass, flooded vegetation, crops, scrub, built area, bare ground, and snow/ice, the coefficients are $[0, 0.10, 0.20, 1.00, 1.00, 0.10, 0.40, 0.15, 0.85, 0]$. After normalization, this gives the guided distribution

$$p_i^{\mathrm{guided}} = \frac{w_i}{\sum_j w_j} \tag{11}$$

This prior increases the chance of sampling patches containing empirically difficult classes in under-covered class–region combinations, while weighting the result by the cached sample-quality score.

We then combine the guided prior with uniform sampling:

$$p_i = (1-\rho) p_i^{\mathrm{uniform}} + \rho p_i^{\mathrm{guided}} \tag{12}$$

Here, $\rho = 1$ corresponds to full guided sampling. We treat this setting as an aggressive ablation. A small ρ yields conservative mixed sampling by retaining a uniform component. We use a fixed empirical mixing coefficient of $\rho = 0.10$ in the core experiments. Uniform sampling uses shuffled mini-batches, whereas mixed guided sampling uses a with-replacement weighted sampler.

For a coverage group G, we quantify its expected training exposure as $M_G = \sum_{i\in G} p_i$, where p_i is the final normalized sampling probability under the strategy being evaluated. Thus, M_G is expected sampling probability mass rather than an observed finite-run sample count.

F. Adaptive Teacher-Guided Soft-Label Fusion

Sampling decides what enters a mini-batch. Fusion decides how non-expert labels are used after sampling. Expert samples retain hard labels. Non-expert samples use adaptive teacher-guided soft-label fusion.

Let y_u^{hard} denote the one-hot non-expert label at pixel u. Let p_u^T denote the raw ten-class teacher softmax output at the same pixel, and let $\bar{p}_u^T$ denote its restriction to the nine valid classes, with $\bar{p}_{u,0}^T = 0$ and $\bar{p}_{u,c}^T = p_{u,c}^T / \sum_{k\in\mathcal{C}} p_{u,k}^T$ for $c \in \mathcal{C}$. We define the fused target as

$$\tilde{y}_u = (1-\lambda_u) y_u^{\mathrm{hard}} + \lambda_u \bar{p}_u^T \tag{13}$$

where $\lambda_u \in [0, 1]$ is the adaptive fusion weight. For adaptive soft-label fusion, the teacher probabilities are restricted to the nine valid classes and renormalized before computing pixel-wise confidence and teacher–label agreement. Thus, fusion confidence is $\bar{\gamma}_{i,u} = \max_{c\in\mathcal{C}} \bar{p}_{i,u,c}^T$, and agreement compares $\arg\max_{c\in\mathcal{C}} \bar{p}_{i,u,c}^T$ with $y_{i,u}$. This differs from the raw ten-class confidence used for sample-level quality estimation. High confidence is defined as $\bar{\gamma}_{i,u} \ge 0.85$, while predictions below this threshold are treated as low confidence. No additional temperature scaling is applied ($T = 1$). The confidence threshold and fusion weights are fixed empirical settings. The fusion weight follows a fixed confidence–agreement rule: high confidence with disagreement uses $\lambda_u = 0.75$; high confidence with agreement uses $\lambda_u = 0.35$; low confidence with agreement uses $\lambda_u = 0.25$; low confidence with disagreement uses $\lambda_u = 0.10$; and invalid pixels use $\lambda_u = 0$. Thus, teacher predictions are used selectively rather than copied unconditionally.

G. Training Objective

For a sampled patch, let $y_{u,c}^*$ be the target distribution: a hard one-hot label for expert pixels and $\tilde{y}_{u,c}$ for valid non-expert pixels. The student uses the dataset-specific priors and weights defined above, with $z'_{u,c} = z_{u,c} - \tau \log \pi_c$. Its objective is

$$\mathcal{L}_{\mathrm{student}} = -\frac{1}{|\Omega_{\mathrm{valid}}|} \sum_{u\in\Omega_{\mathrm{valid}}} \sum_{c=1}^{9} w_c y_{u,c}^* \log \mathrm{softmax}(z'_u)_c. \tag{14}$$

Thus, each component of a non-expert soft target is multiplied by its own class weight before summation; the denominator is the number of valid pixels, not a weighted target sum. Hard-label training is the one-hot special case. The adjusted logits are used for teacher and student training losses only. Teacher probabilities for cache statistics and online fusion, and student inference by raw-logit argmax, do not use logit adjustment.

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TABLE II
DOFA CLASS PRIORS AND CLIPPED INVERSE-FREQUENCY WEIGHTS AFTER DATASET FILTERING. VALUES ARE ORDERED AS WATER, TREES, GRASS, FLOODED VEGETATION, CROPS, SCRUB, BUILT AREA, BARE GROUND, AND SNOW/ICE.

Quantity	Values (valid classes in the stated order)
Expert-only teacher π_c	0.092777, 0.301258, 0.040614, 0.029933, 0.130352, 0.233067, 0.050754, 0.090333, 0.030912
Expert-only teacher w_c	3.247135, 1, 7.417625, 10, 2.311105, 1.292580, 5.935648, 3.334968, 9.745792
Train-all student π_c	0.077687, 0.353859, 0.023135, 0.021826, 0.119363, 0.299945, 0.044716, 0.031354, 0.028116
Train-all student w_c	4.554951, 1, 10, 10, 2.964568, 1.179748, 7.913452, 10, 10

H. Implementation Details

Each backbone has its own expert-only teacher, frozen reference cache, and student initialized from the corresponding teacher checkpoint. The method-level pipeline and 10-epoch teacher plus 10-epoch student schedule are shared, but input adapters, encoders, decoders, batch sizes, and learning-rate parameter groups are backbone-specific. No online teacher or cache update is used.

For the DOFA implementation, the valid-class order is water, trees, grass, flooded vegetation, crops, scrub, built area, bare ground, and snow/ice. The exact loader-filtered priors and clipped weights are listed in Table II; class 0 is ignore with $\pi_0 = 1$ for the adjustment and $w_0 = 0$.

IV. EXPERIMENTS

In this section, we evaluate the proposed framework on a Dynamic World-based semantic segmentation task. We first introduce the experimental setup. We then report the main comparisons, matched-seed DOFA analysis, supervision and reliability analyses, geographic exposure and regional performance, prior and expert-fraction sensitivity, spatial consistency, class-wise analysis, and qualitative results.

A. Experimental Setup

We evaluate the proposed framework on a Dynamic World-based remote sensing semantic segmentation task with nine valid semantic classes. The input consists of six-band remote sensing imagery. The training metadata CSV contains 24,077 rows. After loader filtering, the expert-only teacher set contains 4,060 expert patches, and the student train-all set contains 24,037 patches (4,060 expert and 19,977 non-expert); the test loader contains 409 patches. The DOFA data loader uses the first six HLS raster bands at central wavelengths 0.493, 0.560, 0.665, 0.865, 1.610, and 2.200 μm, scaled by 1/10,000 and clipped to $[0, 1]$. The loader preserves native raster dimensions, and model outputs are bilinearly resized to label-map dimensions when needed. The DOFA protocol uses no augmentation. The official Dynamic World test set released by the dataset provider is used for final evaluation. Dynamic World is adopted in this work not only because it provides mixed-quality annotations, but also because it is a global land-cover product, which is consistent with our target setting of global-scale remote sensing segmentation.

We evaluate eleven backbones: CARL ViT-B [31], THOR-Tiny [25], Copernicus-FM ViT-B [26], Prithvi-EO-2.0 + U-

TABLE III
MAIN COMPARISON ON THE TEST SET. DOFA RESULTS ARE REPORTED AS MEAN $\pm$ STANDARD DEVIATION OVER MATCHED SEEDS 42, 44, AND 3407. GOOGLE DYNAMIC WORLD IS INCLUDED AS THE REFERENCE LAND-COVER PRODUCT.

Method	Training strategy	mIoU (%)
Google Dynamic World	–	44.77
DOFA	Uniform	49.21 ± 1.04
DOFA	Mix10 ($\rho = 0.10$)	48.94 ± 0.55
DOFA	Uniform + Adaptive Soft	50.12 ± 0.40
DOFA	Mix10 ($\rho = 0.10$) + Adaptive Soft	50.97 ± 1.03

TABLE IV
CROSS-BACKBONE MIOU (%) COMPARISON UNDER DIFFERENT SAMPLING AND SOFT-LABEL STRATEGIES.

Model	Uniform	Class-bal.	RFS	SRMF	Entropy	Full	Mix10	Mix20	Uniform+Soft	Mix10+Soft	Mix20+Soft	Best
CARL ViT-B	**43.79**	42.28	43.36	41.51	42.64	39.44	43.04	42.09	43.53	42.13	42.29	Uniform
THOR-Tiny	48.85	48.48	48.25	47.81	47.15	48.32	48.16	48.92	48.84	**48.94**	48.24	Mix10+Soft
Copernicus-FM ViT-B	**29.25**	28.66	28.91	27.25	27.49	28.11	27.96	23.31	27.21	26.61	27.24	Uniform
Prithvi-EO-2.0 + U-Net	47.72	46.78	46.51	47.32	46.45	45.06	47.08	47.16	45.60	46.70	**48.21**	Mix20+Soft
SelectiveMAE ViT-B	32.56	33.17	33.96	33.88	32.42	33.10	32.87	33.11	33.57	33.09	**34.22**	Mix20+Soft
SigLIP2 ViT-B/16-384	31.10	33.27	**34.61**	31.68	21.71	26.57	28.03	31.93	28.15	32.59	32.02	RFS
SMARTIES ViT-B	42.37	39.71	41.16	39.95	**43.77**	40.18	40.60	42.74	42.09	41.95	43.13	Entropy
TerraFM-B	46.29	46.25	44.65	44.07	44.97	44.01	43.80	44.86	**46.92**	45.88	46.23	Uniform+Soft
DOFA ViT-B	49.91	49.14	49.13	47.13	49.77	47.19	47.71	47.60	50.96	**51.12**	49.22	Mix10+Soft
GeoRSCLIP ViT-B/32	26.64	28.27	27.56	28.17	27.39	26.80	27.31	27.70	**28.64**	27.19	27.74	Uniform+Soft
ConvNeXtV2-Tiny	41.90	41.21	42.67	42.15	40.62	40.27	**42.70**	40.24	41.26	41.78	42.59	Mix10

Note: Class-bal. denotes class-balanced sampling. RFS denotes repeat-factor-style sampling. SRMF denotes rare-class and semantic-rich sampling. Entropy denotes entropy-aware sampling. Full denotes full guided sampling with $\rho = 1$. Mix10 and Mix20 denote mixed sampling with $\rho = 0.10$ and $\rho = 0.20$. Soft denotes adaptive teacher-guided soft-label fusion.

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Net [19], [23], SelectiveMAE ViT-B [28], SigLIP2 ViT-B/16 [30], SMARTIES ViT-B [27], TerraFM-B [24], DOFA ViT-B [22], GeoRSCLIP ViT-B/32 [32], and ConvNeXtV2-Tiny [29]. This set covers remote-sensing foundation models, spectral encoders, vision-language backbones, and a convolutional baseline. We use this backbone sweep to test whether the proposed training strategy is backbone-dependent on Dynamic World [4].

We use mean Intersection over Union (mIoU) as the primary metric. The DOFA experimental configuration uses AdamW (weight decay 0.05) with cosine annealing for 10 teacher and 10 student epochs. Teacher backbone and head learning rates are 1×10^{-5} and 1×10^{-4}, respectively; the student learning rate is 1×10^{-4}. Batch size is 12, and the final epoch checkpoint is retained without validation-best selection. These DOFA-specific settings are not generalized to all backbones: CARL and THOR use batch size 4, DOFA and Prithvi use 12, and most other runners use 8, with model-specific input adapters and segmentation heads.

The core fixed-teacher DOFA student analyses use seeds 42, 44, and 3407 and report mean $\pm$ sample standard deviation. Historical cross-backbone and detailed ablation matrices are single-run results whose original run seeds are not fully retained; they are reported as exploratory results and are not equated with later repeated-seed means. The expert-fraction analysis uses teacher seeds 11, 22, and 33 with student seed 42 in each run.

For teacher-calibration analysis, we partition prediction confidence into 15 equal-width bins over $[0, 1]$ and report the bin-wise mean confidence and empirical prediction accuracy in the reliability diagrams.

For the controlled cross-backbone mixing-strength analysis, each backbone is evaluated at $\rho \in \{0, 0.10, 0.15, 0.20\}$ over the matched seeds 42, 44, and 3407 using its corresponding frozen teacher and reference cache while preserving the same within-backbone teacher and initialization protocol. Only the guided-sampling mixing coefficient ρ is changed.

B. Main Comparison

Table III reports the main comparison on the test set. Google Dynamic World obtains 44.77% mIoU. Over the three matched seeds, the DOFA uniform-sampling baseline achieves $(49.21 \pm 1.04)\%$ mIoU. Guided mixed sampling alone gives $(48.94 \pm 0.55)\%$, while adaptive soft-label fusion under uniform sampling increases the mean mIoU to $(50.12 \pm 0.40)\%$. Combining guided sampling with adaptive soft-label fusion gives the highest mean among the four core DOFA configurations, at $(50.97 \pm 1.03)\%$. This corresponds to an average improvement of approximately 1.75 percentage points over DOFA Uniform and 6.20 percentage points over Google Dynamic World.

The matched-seed 2×2 comparison separates the main effects. Hard sampling changes Mix10$-$Uniform by -0.277 percentage points, whereas adaptive soft supervision under uniform sampling changes Uniform+Soft$-$Uniform by $+0.903$ percentage points. Conditional on soft supervision, Mix10 adds $+0.852$ percentage points over Uniform+Soft. The difference-in-differences is therefore $[(50.9690 - 50.1172) - (48.9374 - 49.2145)] = +1.129$ percentage points, with paired standard deviation 0.829 percentage points and a 95% confidence interval of $[-0.931, +3.189]$ percentage points across the three matched student seeds. This is a positive average descriptive interaction, not statistically established synergy; the interval crosses zero at $n = 3$.

C. Cross-Backbone Evaluation

Table IV reports the cross-backbone mIoU matrix, with one training run for each configuration. Full guided sampling is not uniformly beneficial, and soft-label fusion helps some backbones but not all.

To further examine sensitivity to guided-sampling strength, Table V reports results across the evaluated backbones. With adaptive soft-label supervision fixed across all settings, $\rho = 0$ gives the highest mean for six backbones, whereas $\rho = 0.10$ gives the highest mean for THOR, SelectiveMAE, and DOFA, and $\rho = 0.20$ gives the highest mean for CARL and GeoRSCLIP. None of the evaluated backbones obtains its highest mean at $\rho = 0.15$. Overall, the response to guided-sampling strength varies across backbones, indicating that no single value of ρ is consistently preferable across the evaluated models.

D. Sampling and Supervision Ablations

1) Detailed DOFA Sampling-Ratio Ablation: We next report the detailed DOFA ablation. Table VI compares the uniform baseline, alternative sampling rules, and mixed-sampling settings with and without adaptive soft-label fusion in a single-run comparison.

Under the single-run comparison, DOFA Uniform reaches 49.91% mIoU, Mix10 reaches 47.71%, and Mix10+Soft reaches 51.12%. Full guided sampling is 2.72 percentage points below Uniform in this comparison, while hard-label mixed sampling is also lower than Uniform.

As an additional repeated-seed control, we combine conventional class-balanced sampling with the same adaptive soft-label supervision. This configuration reaches $(49.53 \pm 0.65)\%$ mIoU over the three matched seeds, below Uniform+Adaptive Soft $(50.12 \pm 0.40)\%$ and the guided-sampling + adaptive-soft configuration $(50.97 \pm 1.03)\%$ on average.

2) Adaptive Supervision and Teacher Reliability: Table VII reports the adaptive fusion-weight-rule ablation on DOFA under fixed guided mixed sampling with $\rho = 0.10$. The combined confidence–agreement fusion rule reaches the highest average mIoU among the tested variants, at $(50.97 \pm 1.03)\%$ over the three matched seeds.

Separately from the fusion-rule ablation, we examine the relative contribution of confidence and agreement in the sample-quality score q_i. As a diagnostic parameterization of Eq. (7), we define

$$q_i(\beta) = \beta q_i^{\text{conf}} + (1 - \beta) a_i. \tag{15}$$

Agreement-only ($\beta = 0$), the symmetric confidence–agreement setting ($\beta = 0.50$), and confidence-only ($\beta = 1$) obtain 50.44%, 49.86%, and 50.58% mIoU, respectively. These

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TABLE V

CROSS-BACKBONE SENSITIVITY TO THE GUIDED-SAMPLING MIXING COEFFICIENT ρ. ALL PERFORMANCE VALUES ARE MIOU (%). RESULTS ARE REPORTED AS MEAN $\pm$ STANDARD DEVIATION OVER MATCHED SEEDS 42, 44, AND 3407. ADAPTIVE SOFT-LABEL SUPERVISION IS KEPT ENABLED FOR ALL SETTINGS IN THIS CONTROLLED ANALYSIS; THEREFORE, $\rho = 0$ CORRESPONDS TO UNIFORM+ADAPTIVE SOFT RATHER THAN THE HARD-LABEL UNIFORM BASELINE. BOLD INDICATES THE HIGHEST OBSERVED MEAN MIOU AMONG THE TESTED ρ SETTINGS FOR EACH BACKBONE.

Backbone	$\rho = 0$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
CARL ViT-B	40.90 ± 1.57	41.79 ± 0.84	41.27 ± 0.78	$\mathbf{42.09 \pm 0.17}$
THOR-Tiny	48.16 ± 0.46	$\mathbf{48.85 \pm 0.50}$	48.63 ± 0.54	48.53 ± 0.26
Copernicus-FM ViT-B	$\mathbf{28.94 \pm 1.13}$	28.01 ± 0.31	28.91 ± 0.95	27.94 ± 0.74
Prithvi-EO-2.0 + U-Net	$\mathbf{47.06 \pm 0.15}$	46.22 ± 0.28	46.68 ± 0.20	46.34 ± 0.81
SelectiveMAE ViT-B	32.62 ± 0.39	$\mathbf{32.66 \pm 0.30}$	32.54 ± 0.33	31.98 ± 0.44
SigLIP2 ViT-B	$\mathbf{32.93 \pm 0.66}$	32.28 ± 2.17	30.81 ± 0.99	30.97 ± 4.04
SMARTIES ViT-B	$\mathbf{43.32 \pm 0.48}$	41.59 ± 1.09	41.64 ± 0.37	42.28 ± 0.21
TerraFM-B	$\mathbf{44.13 \pm 0.75}$	43.49 ± 0.20	43.61 ± 0.22	43.97 ± 1.12
DOFA ViT-B	50.12 ± 0.40	$\mathbf{50.97 \pm 1.03}$	49.91 ± 0.09	49.86 ± 0.73
GeoRSCLIP ViT-B/32	28.05 ± 0.36	27.56 ± 0.63	27.61 ± 0.60	$\mathbf{28.15 \pm 0.40}$
ConvNeXtV2-Tiny	$\mathbf{42.10 \pm 0.36}$	41.96 ± 0.50	41.74 ± 0.45	41.33 ± 0.50

TABLE VI

DETAILED DOFA ABLATION OF SAMPLING RATIO AND ADAPTIVE SOFT-LABEL FUSION.

Configuration	mIoU (%)	Difference (percentage points) over Uniform
Uniform	49.91	0.00
Class-balanced	49.14	-0.77
RFS	49.13	-0.79
SRMF	47.13	-2.79
Entropy-aware	49.77	-0.15
Full guided	47.19	-2.72
Mix05	49.84	-0.07
Mix10	47.71	-2.20
Mix15	46.33	-3.59
Mix20	47.60	-2.31
Mix25	48.93	-0.98
Uniform+Soft	50.96	+1.05
Mix10+Soft	**51.12**	**+1.21**
Mix15+Soft	49.42	-0.49
Mix20+Soft	49.22	-0.69

Note: Full denotes full guided sampling with $\rho = 1$. Mix05, Mix10, Mix15, Mix20, and Mix25 denote mixed sampling with $\rho = 0.05$, 0.10, 0.15, 0.20, and 0.25. Soft denotes adaptive teacher-guided soft-label fusion.

TABLE VII

ABLATION OF THE ADAPTIVE SOFT-LABEL FUSION RULE ON DOFA. ALL VARIANTS USE GUIDED MIXED SAMPLING WITH $\rho = 0.10$. RESULTS ARE REPORTED AS MEAN $\pm$ STANDARD DEVIATION OVER MATCHED SEEDS 42, 44, AND 3407.

Adaptive fusion variant	mIoU (%)
Hard supervision	48.94 ± 0.55
Fixed $\lambda = 0.50$	50.40 ± 0.77
Confidence-based adaptive fusion	50.60 ± 0.90
Agreement-based adaptive fusion	50.68 ± 0.05
Combined confidence–agreement fusion	50.97 ± 1.03

are single-setting diagnostics and do not identify $\beta = 0.50$ as an optimal setting.

The frozen teacher obtains 46.50% mIoU, 67.07% OA, 9.77% ECE, and a Brier score of 0.46, indicating that the teacher is informative but imperfect. Its reliability varies substantially across categories. Per-class IoU ranges from 9.00% for Grass to 75.24% for Water, while recall ranges from 19.55% to 87.44%. Water and Built Area have relatively low ECE values of 2.12% and 2.47%, whereas Grass, Bare Ground, and Flooded Vegetation exhibit substantially larger ECE values of 47.62%, 27.07%, and 23.90%, respectively. Teacher reliability also varies geographically across the 14 biome groups, with mIoU ranging from approximately 33.38% to 50.30% and ECE from 3.27% to 19.94%. As shown in Fig. 3(a), empirical accuracy generally increases with teacher confidence, confirming that confidence is informative, although the deviation from the ideal calibration line in several populated bins indicates imperfect calibration. Fig. 3(b) further shows substantial class-dependent calibration behavior, while Fig. 3(c) demonstrates geographic variation in calibration across biome groups. These observations support selective teacher guidance rather than unconditional replacement of non-expert labels.

E. Geographic and Sensitivity Analyses

1) Geographic Exposure and Regional Performance: The strict geographic control compares the full sampler with the same sampler after setting only $g_c(x_i) = 1$, while retaining the class prior, quality term, $\rho = 0.10$, soft targets, teacher/cache, and runner protocol. Across student seeds 42, 44, and 3407, Full-minus-NoGeo mIoU differences are -0.606, $+4.290$, and $+2.219$ percentage points, for a paired mean of $+1.968 \pm 2.458$ percentage points; one of three seeds is negative. The sampler changes expected exposure (the probability-vector L_1 difference is 0.03038), but the accuracy effect is variable. This paired test isolates the geographic factor more directly than Geo-only versus Uniform.

Using the training-set median coverage threshold defined in Section III-E, the test set contains 242 under-covered and 167 well-covered samples. In the strict Full-versus-NoGeo comparison, under-covered mIoU changes by $+1.672 \pm 3.104$ percentage points on average, while well-covered mIoU changes by $+2.926 \pm 2.379$ percentage points. The larger mean change on well-covered samples cautions against interpreting the geographic term as a demonstrated under-covered-region accuracy advantage. Biome-level paired means improve in 10 of 14 groups and decline in four. These groups are partitions of the same test set, not spatially held-out regions; the experiment therefore does not establish generalization to unseen geographic areas.

2) Sensitivity to the Empirical Class-Difficulty Prior: To vary the strength of the empirical class-difficulty prior, we

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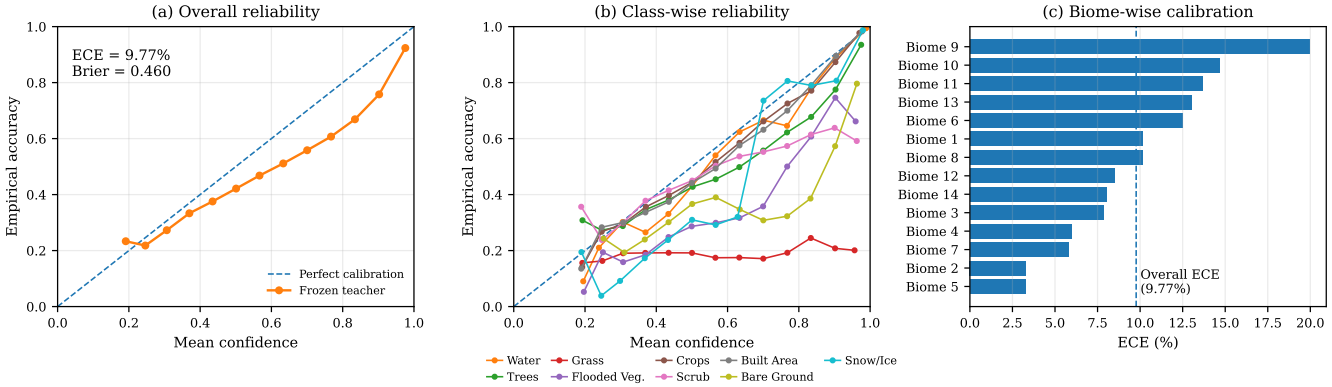


Fig. 3. Frozen-teacher reliability and calibration diagnostics. (a) Overall reliability diagram, where the dashed diagonal denotes perfect calibration. (b) Class-wise reliability curves for the nine valid land-cover classes. (c) Biome-wise Expected Calibration Error (ECE) across the 14 biome groups. The results show that teacher confidence is informative, while calibration quality varies across both classes and geographic regions.

replace α_c with α_c^γ, where $\gamma = 1$ recovers the reference prior, $\gamma < 1$ compresses the differences among nonzero class-difficulty coefficients, and $\gamma > 1$ sharpens them. Sensitivity to this prior remains moderate. Across the three matched seeds, $\gamma = 0.50$, 1, and 2 obtain $(50.23 \pm 1.54)\%$, $(50.97 \pm 1.03)\%$, and $(49.92 \pm 0.94)\%$ mIoU, respectively. Because exponentiation leaves a zero coefficient at zero, these tests do not evaluate whether the Snow/Ice exclusion is appropriate. In separate same-teacher/cache controls, the full prior obtains $(50.9690 \pm 1.0304)\%$, a uniform non-ignore prior ($\alpha_c = 1$ for classes 1–9) obtains $(49.5973 \pm 0.5345)\%$, and setting Snow/Ice α_9 from 0 to 0.10 obtains $(50.5925 \pm 0.2138)\%$ (student seeds 42, 44, and 3407). These small-sample comparisons do not identify a uniquely optimal prior. Relative to the reference setting $\gamma = 1$, $\gamma = 0.50$ decreases the expected exposure of Grass, Flooded Vegetation, and Bare Ground by 3.73%, 4.87%, and 5.94%, whereas $\gamma = 2$ increases their expected exposure by 6.72%, 8.56%, and 7.63%, respectively. These exposure changes are relative changes in expected class-area exposure, not pixel frequencies or final predicted proportions.

3) Expert-Fraction Sensitivity: For expert-only teacher training fractions of 25%, 50%, and 100%, teacher mIoU is $(41.07 \pm 1.18)\%$, $(44.41 \pm 0.96)\%$, and $(46.1164 \pm 0.2415)\%$, respectively; paired students reach $(47.65 \pm 0.13)\%$, $(49.46 \pm 1.55)\%$, and $(50.3504 \pm 0.3117)\%$. The repeated 100% means use teacher seeds 11, 22, and 33, with student seed fixed at 42, matching the 25% and 50% protocol. In all conditions, the student still trains on the full train-all set, including all expert hard labels. This is therefore a teacher-training fraction sensitivity analysis, not an end-to-end annotation-budget experiment. At 100%, the expert subset is fixed, so variation reflects teacher training rather than subset composition.

F. Spatial and Class-Wise Diagnostics

1) Spatial Consistency and Class Adjacency: The seed-42 spatial consistency and class-adjacency diagnostic uses a one-pixel tolerance for Boundary F1. Across the four configurations, Boundary F1 remains within a narrow range of 9.17–9.53%. Relative to the corresponding hard-label settings, adaptive soft supervision reduces Adjacency L1 from 0.20 to 0.19 under uniform sampling and from 0.21 to 0.19 under Mix10; Adjacency Cosine changes from 0.988926 to 0.990037 and from 0.988791 to 0.989853, respectively. Thus, soft supervision improves adjacency-statistics agreement, while Boundary F1 remains nearly unchanged and the guided sampler shows only small or mixed structural effects.

2) Class-wise Analysis on DOFA: Table IX reports a single-run class-wise IoU comparison. Relative to DOFA Uniform, DOFA Mix10+Soft improves seven of the nine valid classes,

TABLE VIII
GEOGRAPHIC EXPOSURE AND CONTROLLED REGIONAL PERFORMANCE EVIDENCE. EXPECTED SAMPLING MASS QUANTIFIES TRAINING EXPOSURE. THE STRICT FULL-VERSUS-NOGEO COMPARISON KEEPS CLASS, QUALITY, SOFT SUPERVISION, TEACHER/CACHE, AND ρ FIXED AND CHANGES ONLY g_c. COVERAGE GROUPS USE THE FIXED TRAINING-SET MEDIAN THRESHOLD $T_{\text{cov}} = 3.41$.

Evidence	Configuration	Under-covered	Well-covered	Overall paired difference (percentage points)
Expected sampling mass	Uniform	0.50	0.50	–
Expected sampling mass	Mix10	0.51	0.49	–
Expected sampling mass	Mix10 without geographic term	0.50	0.50	–
Strict Full mIoU (%), 3 seeds	Full sampler	49.48 ± 1.44	50.76 ± 0.75	–
Strict NoGeo mIoU (%), 3 seeds	$g_c = 1$	47.81 ± 2.27	47.83 ± 1.76	–
Paired Full−NoGeo	Difference (percentage points)	$+1.67 \pm 3.10$	$+2.93 \pm 2.38$	$+1.97 \pm 2.46$

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TABLE IX
CLASS-WISE IOU (%) COMPARISON FOR THE DOFA-BASED DIAGNOSTIC ANALYSIS.

Class	Google DW	DOFA Uniform	DOFA Uniform+Soft	DOFA Mix10+Soft	Difference (percentage points) Mix10+Soft vs Uniform
Water	61.71	76.65	**78.91**	78.17	+1.52
Trees	61.82	64.11	63.43	**64.46**	+0.35
Grass	15.47	8.06	**9.48**	7.44	-0.63
Flooded Vegetation	30.13	38.45	40.19	**42.55**	+4.10
Crops	48.57	56.36	57.29	**58.51**	+2.16
Scrub	32.72	47.25	46.76	**47.62**	+0.37
Built Area	70.27	65.37	66.84	**67.42**	+2.06
Bare Ground	21.19	20.84	**25.31**	22.27	+1.43
Snow/Ice	61.06	**72.13**	70.44	71.63	-0.50

Note: Google DW class-wise IoU is reported under the same evaluation protocol as the main Google DW baseline.

with the largest increase on Flooded Vegetation (+4.10 percentage points). Grass and Snow/Ice decrease by 0.63 and 0.50 percentage points, respectively. These class-wise results motivate the matched-seed Grass analysis below.

Because Grass shows a lower IoU in the class-wise diagnostic, we further examine its behavior over the three matched seeds. Grass IoU is $(8.23 \pm 0.81)\%$, $(9.36 \pm 2.51)\%$, $(12.04 \pm 1.01)\%$, and $(10.36 \pm 3.70)\%$ for Uniform, Mix10, Uniform+Soft, and Mix10+Soft, respectively. Uniform+Soft improves Grass IoU over Uniform in all three runs. In contrast, adding guided sampling under adaptive soft supervision changes Grass IoU by -2.30, -5.24, and $+2.52$ percentage points for seeds 42, 44, and 3407, respectively, corresponding to an average difference of -1.68 percentage points. A more consistent pattern is observed for precision, which is lower for Mix10+Soft than for Uniform+Soft in all three seeds, whereas the recall response changes in both directions. Uniform+Soft also has the highest mean Grass precision and recall among the four configurations. The confusion matrices show that Grass errors are concentrated mainly among Scrub, Crops, and Trees.

G. Qualitative Analysis

Figure 4 provides a qualitative comparison under the DOFA settings listed in the corresponding tables. The examples focus on difficult or ambiguous scenes; fine boundaries and mixed land-cover areas remain difficult.

We further examine learned feature organization in Figure 5, which provides a qualitative comparison of the trained DOFA and Copernicus-FM models. For each backbone, we extract backbone-native feature tokens at the corresponding native feature resolution and retain only tokens with at least 0.80 class purity and 0.80 valid-pixel coverage. Class-aware balanced sampling is used for visualization under the same filtering criteria. The resulting t-SNE panels show different class-wise representation organization and overlap patterns across the two trained models; some groups appear relatively compact, while others remain substantially overlapping. Because the embeddings are qualitative and backbone-specific, absolute t-SNE distances are not directly comparable.

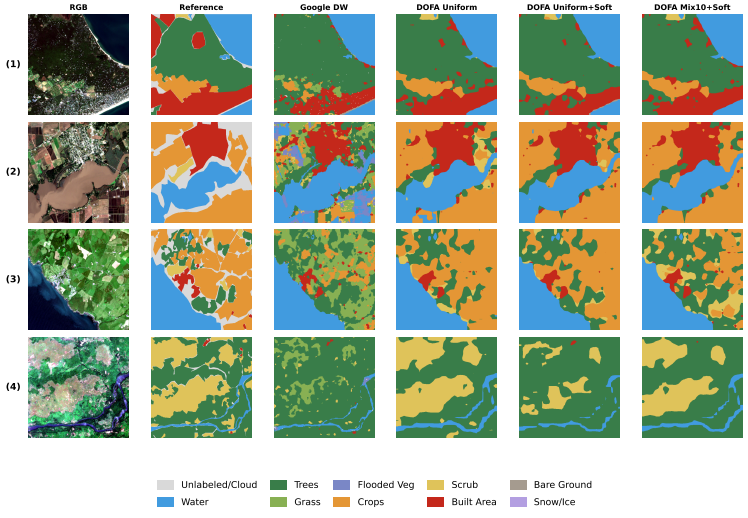


Fig. 4. Final DOFA-based qualitative comparison on representative scenes. The columns show the input image, reference label, Google Dynamic World, DOFA Uniform, DOFA Uniform+Soft, and DOFA Mix10+Soft.

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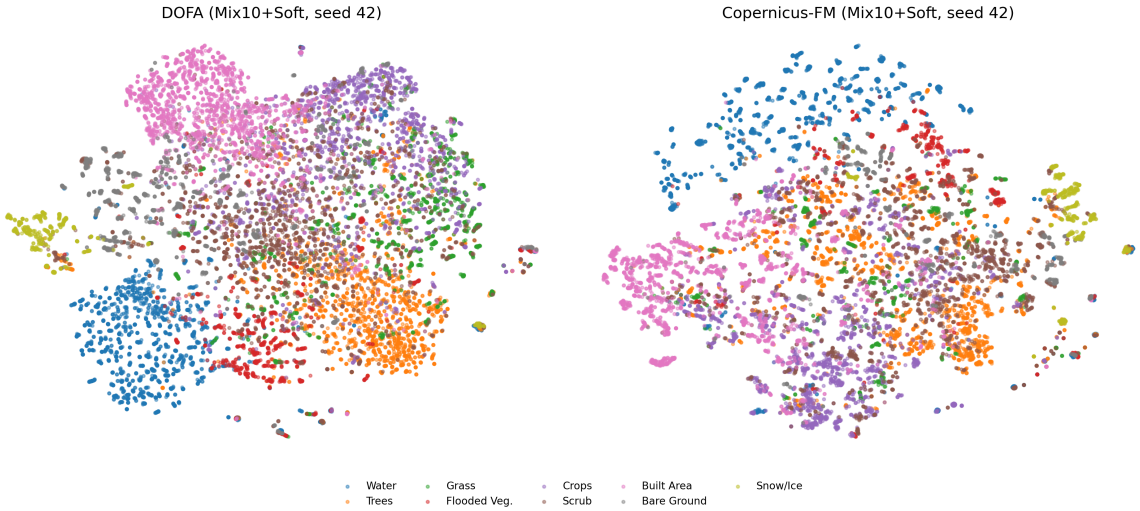


Fig. 5. Feature-token visualization for (a) DOFA and (b) Copernicus-FM. The trained backbone-specific models use their own native feature tokens and native feature resolutions. Tokens satisfying the same class-purity threshold of 0.80 and valid-pixel coverage threshold of 0.80 are retained using class-aware balanced sampling and visualized with t-SNE. The visualization is qualitative and is intended to illustrate differences in class-wise representation organization rather than to provide a direct quantitative comparison of embedding distances.

V. Discussion

The final experiments provide evidence about backbone-dependent performance, the balance between guided and uniform sampling, the role of adaptive soft-label fusion, and the limits of transferring one training strategy across models. Taken together, these results support a conditional conclusion rather than a universal conclusion that holds for all backbones.

A. Backbone Representation and the Strongest DOFA Result

Table V is the main repeated evidence for backbone-dependent behavior because it fixes adaptive soft supervision and varies only ρ over three matched seeds. At $\rho = 0$ (Uniform+AdaptiveSoft), DOFA rises from 50.12% to 50.97% at $\rho = .10$ and THOR rises from 48.16% to 48.85%, while CARL rises from 40.90% to 42.09% only at $\rho = .20$. In contrast, Copernicus-FM falls from 28.94% at $\rho = 0$ to 28.01% at $\rho = .10$, Prithvi falls from 47.06% to 46.22%, TerraFM falls from 44.13% to 43.49%, and ConvNeXtV2 changes from 42.10% to 41.96%. These results do not support a rule that a higher baseline or a particular backbone family must benefit from guided sampling.

The model implementations differ in their spectral adapters and segmentation heads. The six-band HLS input is passed natively with wavelength metadata in Copernicus-FM and through a wavelength-aware native path in CARL; DOFA and Prithvi use batch size 12, CARL and THOR use batch size 4, and the other runners generally use batch size 8. ConvNeXtV2 and GeoRSCLIP/SelectiveMAE use learnable 6-to-3 adapters, TerraFM uses a 6-to-12 adapter, and Prithvi uses a TerraTorch U-Net decoder; SimpleSegHead is used for CARL and THOR, whereas Copernicus-FM and TerraFM use multi-scale heads. These adapters and heads change which spectral information and spatial features reach the pixel classifier. Copernicus-FM nevertheless does not improve at the tested guided ratios, so spectral compatibility is not a sufficient condition for a positive sampling response.

Pretraining objectives also differ: Earth-observation multispectral or spectral-aware pretraining, temporal or multisensor objectives, and RGB or vision-language objectives are not interchangeable with the present single-time, six-band, pixel-dense task [22]–[26], [29], [30]. Token size, patch downsampling, and decoder aggregation may also influence spatial context. These explanations remain hypotheses: effective receptive fields were not measured, and the qualitative, backbone-specific t-SNE visualization and Table IV do not establish causal mechanisms.

The expert-only teacher is another shared dependency: it determines both sample-quality exposure and pixel-wise soft correction, so final student mIoU cannot be used as a proxy for teacher calibration.

B. Full Guided Sampling and Mixed Sampling

The sampling results show that full guided sampling is not stable. In the detailed single-run ablation, full guided sampling reaches 47.19% mIoU, while the DOFA uniform baseline in the same comparison reaches 49.91%. This shows that even if the guided distribution contains class difficulty, geographic under-coverage, and sample quality information, sampling completely according to the guided distribution can still hurt the model's final performance.

One possible reason is that full guided sampling changes the training distribution. Guided sampling increases the probability of sampling difficult classes, geographically under-covered regions, and high-quality samples. In principle, this helps concentrate training resources on more valuable samples. However, if this adjustment is too strong, the model may see some difficult or rare samples too often, while seeing fewer ordinary samples and less of the overall data distribution. For semantic segmentation, over-emphasizing difficult samples

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may hurt the learning of common classes, background regions, and normal spatial combinations.

The role of mixed sampling is to keep a compromise between the guided distribution and the uniform distribution. Uniform sampling provides stable data coverage. Guided sampling provides task-related training priority. After the two distributions are mixed, the model does not completely ignore class difficulty and geographic under-coverage, but it also does not completely deviate from the original training distribution. This is why we do not treat full guided sampling as the final method, but use it as an ablation. The final results show that sampling should be used carefully. It can help organize training data, but an overly strong sampling bias can cause distribution shift.

C. Adaptive Soft-Label Fusion and Hard-Label Training

The matched factorial evidence is reported in Section IV-B; it shows a negative hard-sampling contrast, a larger soft-supervision contrast under uniform exposure, and a positive but uncertain conditional sampling contrast. The interaction is therefore interpreted as descriptive rather than as statistically established synergy. Mix10 changes exposure while leaving hard non-expert targets unchanged.

In contrast, adaptive soft-label fusion under uniform sampling improves the mean mIoU relative to the uniform baseline. This shows that even without changing the sampling distribution, changing how non-expert labels are used can bring a clear improvement. Adaptive soft-label fusion uses teacher predictions to smooth and correct non-expert hard labels. The supervision signal no longer depends completely on a single hard label. For pixels near class boundaries, mixed land-cover regions, or unstable non-expert annotations, the teacher soft prediction can provide extra information, such as class similarity and pixel-level uncertainty.

The results are consistent with the possibility that increasing exposure to difficult or under-covered samples is less effective when non-expert supervision remains hard. Under adaptive soft supervision, Mix10 provides a positive average additional effect. Sampling decides which samples the model sees more often, while soft-label fusion decides how strongly non-expert labels supervise the model after they enter a mini-batch.

This result also shows that the core of the proposed method is not simply to sample more difficult samples. If non-expert labels are still used as hard labels, increasing difficult-sample exposure may make the model more sensitive to annotation noise. Adaptive soft-label fusion changes the supervision signal in the same training pipeline, so Mix10+Soft combines sample reordering with teacher-guided soft targets.

The class-wise results also support this explanation. Compared with DOFA Uniform, Mix10+Soft improves seven classes. Flooded Vegetation, Crops, Built Area, Water, and Bare Ground show clear gains. This shows that the combination of soft-label fusion and mixed sampling improves several difficult or ambiguous categories. However, Grass and Snow/Ice still decrease. This means that the method improves the overall result, but it does not solve all class-specific errors. For some classes, the teacher may still not provide sufficiently reliable correction, or the class itself may have strong mixed appearance and boundary uncertainty in the images.

The Grass analysis illustrates that the average segmentation improvement is not uniform across classes. Although adaptive soft supervision improves Grass under uniform sampling in all three matched seeds, the additional effect of guided sampling is seed dependent and is accompanied by lower Grass precision. The frozen teacher itself performs poorly on Grass, with 9.00% IoU, 19.55% recall, 47.62% ECE, and a Brier score of 1.17. In the seed-42 spatial diagnostic, Grass–Scrub adjacency accounts for 12.15% of the reference Grass adjacency distribution, compared with 32.67% for Uniform+Soft and 36.09% for Mix10+Soft. Together with the concentration of errors among Scrub, Crops, and Trees, these observations are consistent with substantial ambiguity among vegetated categories. The improved aggregate adjacency agreement under soft supervision does not imply improvement for every class.

D. Geographic Exposure and Regional Responses

The geographic term is best interpreted as a structured exposure-allocation mechanism. Mix10 increases the expected sampling probability mass assigned to under-covered training patches from 0.50 to 0.51, whereas the Geo-only control does not independently improve mean mIoU over Uniform. Under adaptive soft supervision, the under-covered and well-covered subsets improve by 0.95 and 1.16 percentage points on average, respectively, while the absolute under-/well-covered performance gap decreases modestly from 1.53 to 1.33 percentage points. Together with the sampling-mass analysis, these results support measurable exposure reallocation and modest average regional changes.

E. Backbone-Dependent Responses to Sampling and Soft-Label Fusion

The single-run configuration matrix complements the repeated-seed Table V. In that table, CARL is a counterexample to a baseline-based rule: its $\rho = 0$ mean is 40.90% and its best tested mean is 42.09% at $\rho = .20$, whereas Prithvi starts higher at 47.06% and decreases to 46.22% at $\rho = .10$. The tested response is therefore a property of the model implementation, teacher, and experimental configuration rather than a monotone consequence of baseline accuracy.

For deployment, the decision should be empirical and resource-aware. First verify the target sensor's band adapter or native wavelength path, the decoder and feature resolution, the Uniform+AdaptiveSoft reference, and teacher calibration or class-wise reliability. Then keep the soft target rule fixed and run the paired ρ sweep at 0, 0.10, and 0.20 within the available memory budget. This procedure makes the added sampler complexity conditional on a measured gain; it does not promote DOFA's $\rho = .10$ to a universal setting.

VI. Conclusion and Future Work

This paper studies remote sensing semantic segmentation with mixed expert and non-expert annotations in the Dynamic World setting. We train an expert-only teacher and use it to

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build a reference cache. Based on this cache, we combine conservative mixed sampling with adaptive teacher-guided soft-label fusion during final training. The sampling branch controls which samples are seen more often, while the soft-label fusion branch controls how strongly non-expert labels are trusted.

In the three-seed core comparison, Uniform reaches $(49.21 \pm 1.04)\%$ mIoU and the guided-sampling plus adaptive-soft configuration reaches $(50.97 \pm 1.03)\%$. Adaptive soft-label fusion provides the more consistent average improvement, while guided sampling provides a positive additional average effect under soft supervision. The class-wise results further show that the combined configuration improves seven of the nine valid classes in the single-run comparison, although Grass and Snow/Ice remain difficult.

The sampler comparison shows that the best strategy depends on the backbone. Full guided sampling can hurt because it may change the training distribution too strongly. Hard-label mixed sampling alone is not consistently beneficial, whereas adaptive soft-label fusion provides the more consistent average improvement. Under soft supervision, guided sampling provides a positive additional average effect. These results support a conditional view.

The geographic component is best interpreted as a structured exposure-allocation mechanism. It increases expected sampling mass for under-covered samples, while the three-seed regional diagnostic shows a positive average response on the under-covered subset and only a modest average reduction in absolute regional disparity. Geo-only does not independently improve mean mIoU.

However, several limitations remain. First, the method depends on the quality of the expert-only teacher. If the teacher is not reliable, the cached confidence scores, sample-quality estimates, and soft predictions may provide weak or even misleading guidance. Second, calibration is important because confidence is used in both sample scoring and soft-label fusion. Poor calibration can weaken both parts of the framework. Third, aggressive guided sampling can introduce training-distribution shift, as shown by the full guided sampling result. Fourth, the current evidence is still limited to the datasets and settings used in this study. Because no independent gold-standard annotation is available for the expert subset, residual annotation errors in the expert labels cannot be separately quantified in the present study. We therefore treat expert annotations as higher-quality anchors rather than assuming that they are noise-free. The regional analysis is based on only three matched seeds and should therefore be interpreted descriptively.

Class-specific analysis further shows that average improvements do not imply uniform gains across classes, with Grass exhibiting seed-dependent behavior and confusion with other vegetated categories.

Transfer beyond optical pixel-level segmentation requires a new task definition. SAR automatic target recognition is commonly image- or object-level recognition rather than pixel-level land-cover segmentation; its teacher distribution and loss would therefore need to be defined at the target level, with exposure priorities tied to target class, acquisition condition, site, or sensor coverage. Confidence calibration must also account for SAR scattering and domain shift. For optical–SAR interpretation, sensor-compatible adapters, cross-modal consistency, or separately calibrated teachers would be required because teacher-label agreement across different observation physics is not equivalent to single-sensor agreement. The SAR review by Zhou et al. emphasizes data quality, evaluation protocols, and domain-aware modeling challenges [43]. SAR and multimodal extensions remain untested in this study. Future work should evaluate the framework on additional datasets, improve reliability estimation, and test different label-noise patterns and geographic distributions.

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